

upGrad | Enterprise Data Science on AWS

Topic 02: Validate before analysis

Data foundations: from S3 objects to analysable tables

M00 Foundations Pre-Module | Topic 02

TOPIC 02

Validate before analysis

Inspect types, missingness, time and entity keys before transformations begin.

Inspect before you transform

```
1 import pandas as pd
2
3 orders = pd.read_csv("data/orders.csv")
4 print(orders.dtypes)
5 print(orders.isna().mean().sort_values(ascending=False).head())
6
7 orders["ordered_at"] = pd.to_datetime(orders["ordered_at"], utc=True)
8 assert orders["order_id"].is_unique
9
```

The first model feature is often a validation check: are we measuring the entity and time period we think we are?

Key takeaway

Key Takeaway – Storage is not semantics

S3 gives durable object storage. Naming conventions, schema checks and documented transformations make those objects safe to analyse.